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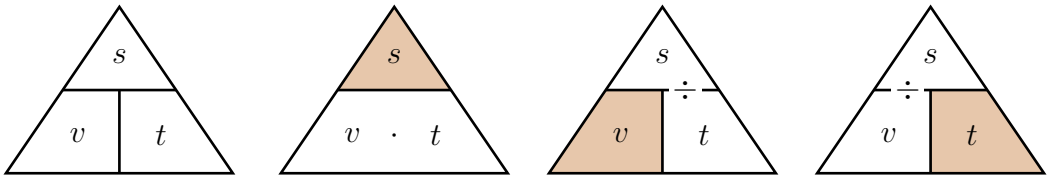
Mr. Rawson • Y8 MYP Physics

Speed Calculations 2: Real Numbers

Unit 2: Forces and Motion • decimals, units and challenges

The triangle, again

Same triangle, same four steps as Sheet 1. Shade the letter you are looking for.



the triangle **shade s:** $s = v \cdot t$ **shade v:** $v = s \div t$ **shade t:** $t = s \div v$

Two new moves

1. Round to 1 decimal place. Real numbers rarely divide exactly. Write the calculator answer, then round it: look at the second decimal place, and if it is 5 or more, round up.

2. Units in, units out. The answer comes out in the units you put in. Put in kilometres and hours, and a speed comes out in km/h; put in miles and miles per hour, and a time comes out in hours.

Worked example — copy this shape

A bus travels **300 m** in **230 s**. What is its speed?

WRITE $s = 300$ m $t = 230$ s $v =?$
CHOOSE shade v: $v = s \div t$
PUT IN $v = 300 \div 230 = 1.304\ldots$
ANSWER $v =$ **1.3 m/s** (to 1 decimal place)

Worked example — copy this shape

A cyclist rides at **20 km/h** for **3 hours**. How far does she ride?

WRITE $v = 20$ km/h $t = 3$ h $s =?$
CHOOSE shade s: $s = v \cdot t$
PUT IN $s = 20 \cdot 3$
ANSWER $s =$ **60 km** (km/h and hours give kilometres)

Check: round once, at the very end, and write the unit after every answer.

Your turn

Use the four steps for every question. Give speeds and times to 1 decimal place.

(1) A student runs **600 m** in **120 s**. How fast is she?

WRITE $s = 600$ m $\quad t = 120$ s $\quad v = ?$

CHOOSE $v = s \div t$

PUT IN $v = 600 \div 120$

ANSWER $v = 5$ **m/s**

(2) A badminton player hits a shuttlecock **50 m** in **2 s**. How fast does it go?

WRITE $s = 50$ m $\quad t = 2$ s $\quad v = ?$

CHOOSE $v = s \div t$

PUT IN $v = 50 \div 2$

ANSWER $v = 25$ **m/s**

(3) A sprinter runs **100 m** in **12.4 s**. What is her speed?

WRITE $s = 100$ m $\quad t = 12.4$ s $\quad v = ?$

CHOOSE $v = s \div t$

PUT IN $v = 100 \div 12.4 = 8.064\ldots$

ANSWER $v = 8.1$ **m/s**

(4) A hockey ball is hit at **40 m/s**. How far does it go in **5 s** at that speed?

WRITE $v = 40$ m/s $\quad t = 5$ s $\quad s = ?$

CHOOSE $s = v \cdot t$

PUT IN $s = 40 \cdot 5$

ANSWER $s = 200$ **m**

(5) A hockey player runs with the ball at **8 m/s**. How long does it take her to run **80 m**?

WRITE $s = 80$ m $\quad v = 8$ m/s $\quad t = ?$

CHOOSE $t = s \div v$

PUT IN $t = 80 \div 8$

ANSWER $t = 10$ **s**

(6) A plane flies at **350 miles per hour**. London to Singapore is **6750 miles**. How long does the flight take?

WRITE $s = 6750$ miles $\quad v = 350$ mph $\quad t = ?$

CHOOSE $t = s \div v$

PUT IN $t = 6750 \div 350 = 19.28\ldots$

ANSWER $t = 19.3$ **hours**

Challenges — to go further

Challenge 1. Two cars are **4 km** apart, driving the same way along a road. The car behind goes at **10 m/s**; the car in front goes at **9 m/s**. How long until the back car catches up, and how far has each car travelled by then?

Hint: the gap shrinks by 1 m every second, and 4 km is 4000 m.

The gap closes at $10 - 9 = 1$ m/s, so $t = 4000 \div 1 = 4000$ s (about 67 minutes). Back car: $s = 10 \cdot 4000 = 40\,000$ m $= 40$ km. Front car: $s = 9 \cdot 4000 = 36\,000$ m $= 36$ km. The difference is the 4 km gap.

Challenge 2. (a) What is **100 km/h** in m/s? (b) What is **100 miles per hour** in m/s?

Hint: 1 km = 1000 m, 1 mile = 1609 m, and 1 hour = 3600 s. Change the distance to metres and the time to seconds, then divide.

(a) $100\,000\ m \div 3600\ s = 27.8\ m/s$. *(b)* $160\,900\ m \div 3600\ s = 44.7\ m/s$.

Challenge 3. Two cars are **20 km** apart, driving towards each other. One goes at **40 km/h** and the other at **80 km/h**. (a) How long until they meet? (b) Where do they meet?

Hint: together they close the gap by $40 + 80 = 120$ *km every hour.*

(a) $t = 20 \div 120 = 0.17$ *hours, which is 10 minutes.* *(b) In that time the slower car drives* $40 \cdot \frac{1}{6} = 6.7$ *km, so they meet 6.7 km from the slower car's start (13.3 km from the faster car's).*